

Publications

1. **B. Assi**, R. Plestid, A. Sengupta, J. Zupan. 2026. “The flavour of $SU(15)$ composite quarks and leptons”. arXiv:2608.00166 [hep-ph].
2. **B. Assi**, A. Kronfeld, S. Vaiva, M.L. Wagman. 2026. “Renormalon subtracted nonrelativistic QCD for heavy hadron systems”. arXiv:2607.08817 [hep-ph].
3. **B. Assi**, A. Martin and W. Shepherd. 2026. “EFT Validity and Truncation Uncertainty from Few Nuisance Parameters”. arXiv:2607.02649 [hep-ph].
4. **B. Assi** and P.A. Machado. 2026. “Neutrino Dipole Moments and Radiative Signatures from Partial Compositeness”. arXiv:2606.20987 [hep-ph].
5. **B. Assi**, C. Bierlich, R. Gambhir, P. Ilten, T. Menzo, S. Mrenna, M. Szewc, M. Wilkinson, J. Zupan. 2026. “Vistas: A Visualization Interface for Particle Collision Simulations”. arXiv:2606.19524 [hep-ph].
6. **B. Assi**, K. Lee and J. Thaler. 2026. “Improving parton shower predictions via precision moments of energy flow polynomials”. arXiv:2604.00084 [hep-ph].
7. **B. Assi**, C. Bierlich, R. Gambhir, P. Ilten, T. Menzo, S. Mrenna, M. Szewc, M. Wilkinson, J. Zupan. 2026. “HD-Sense: An efficient method for ranking observable sensitivity”. arXiv:2602.01509 [hep-ph].
8. **B. Assi**, S. Carey, G. Lee, G. Paz, G. Perez, J. Zupan. 2025. “Effective Field Theory Perspective On King Non-linearity”. arXiv:2512.03157 [hep-ph].
9. A. Butter, A. Ore, S. Palacios Schweitzer, T. Plehn, **B. Assi**, C. Bierlich, P. Ilten, T. Menzo, S. Mrenna, M. Szewc, M.K. Wilkinson, A. Youssef and J. Zupan. 2026. “Iterative HOMER with uncertainties”. *SciPost Phys.* *20*, 042. arXiv:2509.03592 [hep-ph].
10. **B. Assi**, A.V. Grebe and M.L. Wagman. 2026. “Baryon-baryon, meson-meson, and meson-baryon interactions in nonrelativistic QCD”. *Phys. Rev. D* *113*, 016013. arXiv:2508.10090 [hep-ph].
11. **B. Assi**, C. Bierlich, P. Ilten, T. Menzo, S. Mrenna, M. Szewc, M. Wilkinson, A. Youssef, J. Zupan. 2025. “Post-hoc reweighting of hadron production in the Lund string model”. *SciPost Phys.* *19*. arXiv:2505.00142 [hep-ph].
12. **B. Assi**, A. Helset, C. Shen, J. Pagès. 2025. “Renormalizing Two-Fermion Operators in the SMEFT via Supergeometry”. *JHEP* *12*. arXiv:2504.18537 [hep-ph].
13. **B. Assi** and A. Martin. 2025. “Energy-Enhanced Expansion of the Standard Model Effective Field Theory”. *Phys.Rev.D* *112*. arXiv:2504.10617 [hep-ph].
14. **B. Assi**, C. Bierlich, P. Ilten, T. Menzo, S. Mrenna, M. Szewc, M. Wilkinson, A. Youssef, J. Zupan. 2025. “Characterizing the hadronization of parton showers using the HOMER method”. *SciPost Phys.* *19*. arXiv:2503.05667 [hep-ph].

15. **B. Assi**, K. Lee, S. Höche, J. Thaler. 2025. “QCD Theory Meets Information Theory”. *Phys.Rev.Lett.* 135. arXiv:2501.17219 [hep-ph].
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18. **B. Assi**, W. Altmannshofer, J. Brod, N. Hamer, J. Julio, P. Uttayarat, D. Volkov. 2024. “Electron EDM and $\Gamma(\mu \rightarrow e\gamma)$ in the 2HDM”. *JHEP* 06. arXiv:2410.17313 [hep-ph].
19. **B. Assi** and H. Lamm. 2024. “Digitization of $SU(N)$ Gauge Theories”. *Phys.Rev.D* 110. arXiv:2405.12204 [hep-ph].
20. **B. Assi** et al. 2024. “Les Houches 2023: Physics at TeV Colliders”. arXiv:2406.00708 [hep-ph].
21. **B. Assi** and M.L. Wagman. 2023. “Tetraquarks made of sufficiently heavy and unequal mass quarks are bound in QCD”. *Phys.Rev.D* 110. arXiv:2311.01498 [hep-ph].
22. **B. Assi**, A. Helset, A.V. Manohar, C. Shen, J. Pagès. 2023. “Fermion Geometry and the Renormalization of the Standard Model Effective Field Theory”. *JHEP* 11. arXiv:2307.03187 [hep-ph].
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27. **B. Assi**, B.A. Kniehl, J. Soto. 2020. “Matching the Standard Model to Heavy Quark Effective Theory and nonrelativistic QCD”. *Nucl.Phys.B* 992. arXiv:2011.06447 [hep-ph].
28. **B. Assi** and B.A. Kniehl. 2020. “Electroweak Form Factor in Sudakov and Threshold Regimes with Effective Field Theories”. arXiv:2011.14933 [hep-ph].
29. **B. Assi**, B.A. Kniehl, O. Veretin. 2020. “Static Potential of the Standard Model and Spontaneously Broken Theories”. arXiv:2011.06437 [hep-ph].
30. **B. Assi**. 2021. *PhD Thesis: Electroweak Radiative Corrections and Effective Field Theories*. (internal publication).
31. **B. Assi**. 2018. *M.Res. Thesis: Modelling the Integer Quantum Hall Effect in Geometrically Disordered Chalker–Coddington Networks*. (internal publication).

32. **B. Assi.** 2017. *M.Sc. Thesis: Causal Set Action and Curvature.* (internal publication).